

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester III Examination, 2025-26 (NEP)

CHEMISTRY**Paper Code: CHEMJ33 : Physical Chemistry-I**

Ref. No.: CECONF/N/2025-26/057

Time: 2½ Hours

Maximum Marks: 70

Instructions: Answer five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1.

- (a) Construct a plot for the conductometric titration of an aqueous mixture of HCl and CH₃COOH solution with an aqueous sodium hydroxide solution and justify it. [3]
- (b) Write about the two types of electrodes and their electrode reactions. [2]
- (c) What is buffer solution? Write the conditions under which it can show maximum buffer-capacity. [1+2]
- (d) Discuss the eutectic point and the triple point with suitable examples. [2+2]
- (e) How can you determine the activation energy of a reaction? [2]

Question 2.

- (a) Define molar conductance and equivalent conductance of an electrolyte. Write their units. [3]
- (b) Explain why proton (H⁺) has a very high molar conductivity in water. [3]
- (c) What is the transference number of ion? Discuss the principle for the determination of the transference number using moving boundary method. [1+3]
- (d) Write the Kohlrausch's law of independent migration of ions. Discuss, with a suitable example, the use of it for the determination of the conductance of a weak electrolyte. [2+2]

Question 3.

- (a) Write the relation between the ionic mobility and its limiting molar conductivity. Justify it. [6]
- (b) Discuss the applications of conductance measurements. [3]
- (c) Write the basic principle for the conductometric titrations. [3]
- (d) Predict the plot for the conductometric titration of NH₄OH vs. HBr. [2]

Question 4.

- (a) What is reference electrode? Discuss with a suitable example. [2]
- (b) Define the following terms: [2+1]

- (i) Cell potential
- (ii) Standard electrode potential
- (c) Write the relation between free energy change and cell potential of an electrochemical cell and justify it. From it, derive the Nernst equation for an electrochemical cell having a general cell reaction: $aA + bB \rightleftharpoons cC + dD$. [3+3]
- (d) Discuss the principle for the determination of the pH of a solution using a glass electrode. [3]

Question 5.

- (a) Considering the following cell: $\text{Hg}(l) \mid \text{Hg}_2\text{Cl}_2(s) \mid \text{KCl (std. soln.)} \parallel \text{Fe}^{+2}(aq.), \text{Fe}^{+3}(aq.) \mid \text{Pt}(s)$ and assuming the redox reaction: $\text{Fe}^{+2}(aq.) + \text{Ce}^{+4}(aq.) = \text{Fe}^{+3}(aq.) + \text{Ce}^{+3}(aq.)$ involving the oxidation of Fe^{+2} ions by Ce^{+4} ions carried out in a potentiometric titration, write the expressions for the E_{cell} before and after the equivalence point (Given: $E_{\text{calomel(satd.)}}^\circ = +0.242 \text{ V}$, $E_{\text{Fe}^{+3}/\text{Fe}^{+2}}^\circ = +0.77 \text{ V}$ and $E_{\text{Ce}^{+4}/\text{Ce}^{+3}}^\circ = +1.61 \text{ V}$ at 298 K). [3]
- (b) Justify the nature of the expected plot to be obtained from the above titration. [3]
- (c) What is acid-base indicator? Derive a relation between pH a solution with the pK_{ind} value of an acid indicator. What is the working pH range for the effective usefulness of an indicator? Justify your answer. [2+2+2]
- (d) Discuss the choice indicators for the titrations of (i) strong acid vs. strong base, (ii) weak acid vs. strong base, (iii) strong acid vs. weak base and (iv) weak acid vs. weak base. [2]

Question 6.

- (a) Derive the Clapeyron-Clausius equation. [3]
- (b) Justify the phase rule. [4]
- (c) Use it to predict the maximum number of phases to be available for a single component system. Predict the position of these phases in the temperature vs. pressure phase diagram. [3]
- (d) Draw the phase diagram of water system at low pressure. Use the phase rule to calculate the no. of degrees of freedom (variance) at the following points (i) within the two curves, (ii) on the curve and (iii) at the junction points of all the curves. [2+2]

Question 7.

- (a) Derive Nernst distribution law. [2]
- (b) Workout the distribution law of a solute undergoing dissociation in one solvent. [2]
- (c) Describe the procedure to determine the equilibrium constant of the reaction: $\text{KI} + \text{I}_2 \rightleftharpoons \text{KI}_3$ in aqueous medium using Nernst distribution law. [4]
- (d) Define the following terms: (i) Rate of reaction (ii) Rate laws of a reaction (iii) Order of a reaction. [3]
- (e) Derive the rate law for a zero-order reaction. Draw a kinetics plot for it. Give one example for such reaction. [1+1+1]

Question 8.

- (a) Derive the integrated rate law for a first-order reaction considering the rate as the formation of the product. Write the characteristics of such reaction. [2+1]
- (b) Derive the integrated rate law for a second-order reaction considering the following equation: $A + B \rightarrow \text{Product}$. [2]
- (c) Workout the rate law for a pseudo first-order reaction. Justify that acid catalyzed inversion of cane sugar or hydrolysis of ester follows a pseudo first-order kinetics. [2+1+2]
- (d) Discuss the application of the steady state approximation concept with a suitable example. [2]
- (e) Write the three main factors affecting the rate of a reaction according to collision theory. [2]
- (f) Write the postulates of the transition state theory. [2]
